VTU-ETR Seat No. | | | | | | |

Question Paper Version : A

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research), January – 2025

DO NOT USE MOBILE PHONES

Computer Science and Engineering / Information Science and Engineering

Time: 3 hrs. Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
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6. There shall be no negative marking for wrong answers.
7. The examinees shall select the response which they want to mark/indicate on the response sheet and shade/darken/blacken completely the inside area of the corresponding checkbox bubble of circular or oval shape.
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Key Answers (CS/IS): **Version A**

PART - I

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
1	**b**	11	**a**	21	**c**	31	**a**	41	**b**
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3	**b**	13	**b**	23	**b**	33	**b**	43	**a**
4	**c**	14	**c**	24	**d**	34	**a**	44	**c**
5	**c**	15	**c**	25	**d**	35	**c**	45	**b**
6	**b**	16	**c**	26	**b**	36	**a**	46	**a**
7	**d**	17	**d**	27	**c**	37	**d**	47	**b**
8	**a**	18	**c**	28	**b**	38	**d**	48	**c**
9	**b**	19	**d**	29	**b**	39	**b**	49	**b**
10	**c**	20	**b**	30	**b**	40	**c**	50	**b**

PART – II

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
51	C	61	D	71	C	81	A	91	B
52	B	62	D	72	D	82	D	92	B
53	C	63	C	73	A	83	D	93	B
54	D	64	C	74	A	84	D	94	C
55	A	65	B	75	C	85	B	95	C
56	C	66	C	76	D	86	C	96	B
57	B	67	B	77	B	87	C	97	D
58	B	68	D	78	A	88	B	98	C
59	A	69	B	79	A	89	B	99	B
60	B	70	B	80	C	90	C	100	D

PART – I

(Research Methodology)

[50 Marks]

1. What is the main objective of applied research?
a) Develop theories for future studies
b) Solve specific, practical problems
c) Investigate unexplored phenomena
d) Study the historical context of an issue

2. Which of the following steps is NOT part of the research process?
a) Defining the research problem
b) Reviewing related literature
c) Avoiding data collection
d) Formulating a hypothesis

3. Which of the following is NOT a characteristic of good research?
a) Systematic and logical
b) Based on subjective interpretation
c) Replicable and verifiable
d) Directed towards problem-solving

4. Why is it necessary to define a research problem clearly?
a) To reduce the scope of the study
b) To formulate irrelevant hypotheses
c) To guide the research process effectively
d) To avoid reviewing related literature

5. Which research approach combines both qualitative and quantitative methods?
a) Descriptive research
b) Experimental research
c) Mixed-method research
d) Longitudinal research

6. What is the primary purpose of a literature review in research?
a) To test hypotheses
b) To identify gaps in existing knowledge
c) To collect primary data
d) To summarize the results

7. Which of the following is NOT a feature of a good research design?
a) Flexibility
b) Objectivity
c) Economical execution
d) Bias in data collection

8. Which of the following is a principle of experimental design?
a) Randomization
b) Subjectivity
c) Hypothesis elimination
d) Literature review

9. What is a conceptual framework?
a) A plan for data analysis
b) A framework to explain concepts and relationships in the research
c) A compilation of all references
d) A step-by-step experimental protocol

10. Which type of research design is best for studying cause-and-effect relationships?
a) Descriptive design
b) Correlational design
c) Experimental design
d) Exploratory design

11. Type-I errors occurs if ________
a) Null Hypothesis is rejected even if it is true
b) Null Hypothesis is accepted even if it is false
c) Both Null hypothesis and alternative hypothesis are rejected
d) Both Null hypothesis and alternative hypothesis are accepted

12. Which is a preferred sampling method for the population of a finite size.
a) Systematic sampling
b) Purposive Sampling
c) Cluster Sampling
d) Area Sampling

13. Which of the following is a probability sampling method?
a) Convenience sampling
b) Stratified random sampling
c) Quota sampling
d) Judgmental sampling

14. Which classification of measurement scale represents ordered categories but does not have a true zero point?
a) Nominal scale b) Ordinal scale
c) Interval scale d) Ratio scale

15. Which method is most suitable for collecting primary data when studying behaviour over time?
a) Case study method
b) Experimental method
c) Longitudinal survey
d) Secondary data collection

16. Which of the following is NOT a step in the data preparation process?
a) Editing b) Coding
c) Data collection d) Transcribing

17. What type of analysis focuses on summarizing the data collected?
a) Inferential analysis
b) Diagnostic analysis
c) Predictive analysis
d) Descriptive analysis

18. Which measure of central tendency is best suited for categorical data?
a) Mean b) Median c) Mode d) Standard deviation

19. What does skewness measure in a dataset?
a) The average value of the data
b) The spread of data points
c) The relationship between variables
d) The symmetry of the data distribution

20. What is the correlation coefficient used to measure?
a) The degree of dispersion in the data
b) The relationship between two variables
c) The central value of a dataset
d) The extent of symmetry in the distribution

21. Which of the following is an example of a non-sampling error?
a) Errors in calculating the standard error
b) Errors due to small sample size
c) Errors due to incomplete responses
d) Errors caused by random sampling

22. What does the Central Limit Theorem state?
a) The sampling distribution of the mean approaches a normal distribution as sample size increases
b) The mean of the population equals the mean of the sample
c) The variance of the population is less than that of the sample
d) The sampling method used always determines the distribution

23. What is the standard error?
a) The difference between the sample mean and population mean
b) The standard deviation of the sampling distribution
c) The sum of all errors in data collection
d) The variance of a data set

24. What is the main goal of statistical inference?
a) To describe the collected data
b) To evaluate the quality of the data
c) To calculate the mean of the sample
d) To make predictions about the population based on the sample

25. What is a sampling distribution?
a) The distribution of the population data
b) The histogram of the sampled data
c) The error associated with a specific sample
d) The distribution of a statistic derived from all possible samples of the same size

26. What does the P-value represent in hypothesis testing?
a) The probability of making a Type II error
b) The probability of observing the data if the null hypothesis is true
c) The critical value for the test statistic
d) The standard deviation of the sample

27. Which test is used to compare the means of two independent samples?
a) Chi-square test
b) Z-test for proportion
c) Two-sample t-test
d) ANOVA

28. What does the chi-square goodness-of-fit test evaluate?
a) The mean difference between two groups
b) Whether observed data follows a specific distribution
c) The equality of variances between two samples
d) The correlation between two variables

29. Which hypothesis test is used to compare the variances of two populations?
a) Z-test for variance
b) F-test
c) Paired t-test
d) Chi-square test for independence

30. What is a null hypothesis (H_0)?
a) A hypothesis that predicts a significant relationship between variables
b) A hypothesis that states there is no significant effect or relationship
c) A hypothesis tested only for large samples
d) A hypothesis that must always be true

31. What does the analysis of variance (anova) primarily test for?
a) Differences in means across multiple groups
b) Differences in medians across two groups
c) Relationships between two variables
d) Equality of variances across groups

32. What is the main difference between one-way and two-way ANOVA?
a) One-way ANOVA tests one group, while two-way ANOVA tests two groups
b) One-way ANOVA examines one independent variable, while two-way ANOVA examines two independent variables
c) Two-way ANOVA is only for experimental designs
d) One-way ANOVA includes interaction effects

33. In a Latin-square design, how are the treatments arranged?
a) Randomly across all rows
b) Such that each treatment appears once in each row and column
c) To maximize variability in the design
d) Only within rows and not columns

34. What is the F-ratio in ANOVA?
a) The ratio of between-group variance to within-group variance
b) The ratio of group means to total variance
c) The sum of squares within groups
d) The mean of all variances

35. Which of the following is NOT an assumption of ANOVA?
a) Homogeneity of variances
b) Independence of observations
c) Equal sample sizes in all groups
d) Normality of residuals

36. Which of the following is a key characteristic of dependent and independent variables in linear regression?
a) Dependent variable is predicted by the independent variable.
b) Both dependent and independent variables are random variables.
c) Independent variables depend on dependent variables.
d) Dependent variables are not affected by independent variables.

37. What problem arises when independent variables in a multiple regression model are highly correlated with each other?
a) Heteroscedasticity
b) Homoscedasticity
c) Autocorrelation
d) Multicollinearity

38. Which of the following is an example of a qualitative explanatory variable in regression analysis?

a) Age of a person
b) Height of an individual
c) Weight of an individual
d) Type of education (e.g., High School, College, Graduate)

39. What is the primary purpose of factor analysis?
a) To predict future values based on historical data.
b) To reduce the number of variables by identifying underlying factors.
c) To determine the causal relationship between variables.
d) To test hypotheses in a regression model.

40. Which of the following is a key difference between R-type and Q-type factor analysis?

a) R-type focuses on factors affecting the variables, while Q-type analyzes the variables' impact on the factors
b) R-type is used when analyzing observations across multiple time periods, while Q-type focuses on cross-sectional data
c) R-type factor analysis is applied to variables, whereas Q-type is applied to cases (individuals)
d) There is no significant difference between R-type and Q-type factor analysis

41. Which of the following best defines a random experiment?

a) A process that results in one of a set of outcomes, with certainty.
b) A process with uncertain outcomes that can be repeated under identical conditions.
c) A deterministic process where outcomes are predictable.
d) A set of outcomes that always yield the same result.

42. The probability of the occurrence of a single event, such as P(A) or P(B), which takes a specific value irrespective of the values of other events, i.e. unconditioned on any other events, is called?

a) Unconditional Probability
b) Marginal Probability
c) Conditional Probability
d) Joint Probability

43. What is the probability mass function (PMF) used to describe?

a) The likelihood of each outcome in a discrete probability distribution.
b) The cumulative probability of outcomes in a continuous distribution.
c) The variance of a continuous random variable.
d) The mean of a discrete random variable.

44. What does the Axiom of Probability P(S) = 1 mean, where S is the sample space?

a) The probability of any event is 0.
b) The probability of the sample space is 0.
c) The probability of the sample space is 1.
d) The probability of any event is 1.

45. Which of the following is true about a random variable?
a) It takes a fixed set of values.
b) It is a function that assigns a numerical value to each outcome in a sample space.
c) It has a deterministic relationship with other random variables.
d) It is not subject to probability distributions.

46. The binomial distribution is used to model which of the following?
a) The number of successes in a fixed number of independent trials with two possible outcomes.
b) The time between events in a fixed interval.
c) The number of occurrences of an event in a fixed interval of time.
d) The number of unique outcomes in a sample space.

47. The normal distribution is characterized by which of the following?
a) A skewed distribution with a peak at the right.
b) A symmetric bell-shaped curve where the mean, median, and mode are equal.
c) A discrete distribution with a fixed number of outcomes.
d) A distribution that models time between events.

48. Under which condition can the binomial distribution be approximated by a normal distribution?
a) When the number of trials is very small
b) When the probability of success is very high
c) When the probability of success is close to 0.5
d) When the trials are dependent

49. The Poisson distribution is typically used to model:
a) The number of successes in a fixed number of trials
b) The number of events occurring in a fixed interval of time or space
c) The time between events in a Poisson process
d) The number of distinct outcomes in a discrete sample

50 The hypergeometric distribution is used to model which of the following?
a) The number of successes in a fixed number of independent trials with two outcomes
b) The number of successes when the probability of success is constant across trials
c) The number of events occurring in a fixed interval of time
d) The number of successes in a fixed number of trials without replacement from a finite population.

* * * * *

PART – II
Discipline Oriented Section
(Computer Science and Engineering / Information Science and Engineering)

[50 MARKS]

51. Predict the output of following program

```
#include <stdio.h>
int fun(int n)
{
   if (n == 4)
      return n;
   else return 2*fun(n+1);
}
int main()
{
  printf("%d", fun(2));
  return 0;
}
```

a) 4 b) 8 c) 16 d) Runtime Error

52. A Binary search tree is generated by inserting the following elements in order,
50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24
The number of nodes in the left sub tree and right sub tree of the root is ________
a) (4, 7) b) (7, 4) c) (8, 3) d) (3, 8)

53. A variant of the linked list in which none of the node contains NULL pointer is______
a) Singly linked list
b) Doubly linked list
c) Circular linked list
d) None

54. If p and q are independent propositions, which of these is always false?

a) $p \wedge \neg q$ b) $\neg p \wedge \neg q$ c) $p \rightarrow q$ d) $p \leftrightarrow \neg q$

55. How many equivalence relations can be defined on a set of size 3?

a) 5 b) 10 c) 15 d) 6

56. In a 7-node directed cyclic graph, the number of Hamiltonian cycle is to be

a) 180 b) 720 c) 360 d) 540

57. What is the purpose of a Use Case Diagram in UML?
a) To model the flow of data within the system
b) To show the system's interactions with external entities
c) To define the class structure of the system
d) To represent the system's behavior over time

58. In the Rational Unified Process (RUP), which phase focuses on creating the architectural foundation of the system?
a) Inception b) Elaboration
c) Construction d) Transition

59. Which type of testing ensures that previously developed and tested software still works after changes?
a) Regression Testing
b) Acceptance Testing
c) Integration Testing
d) System Testing

60. Which of the following acts as a temporary storage location to hold an intermediate results for mathematical and logical calculations
a) RAM
b) Accumulator
c) Program counter
d) Memory Address Register

61. The unit which acts as an intermediate agent between memory and backing store to reduce process time is ___________
a) Translation Lookaside Buffer's
b) Registers
c) Page Tables
d) Cache

62. The bit used to signify that the cache location is updated is ________
a) Flag bit b) Reference bit
c) Update bit d) Dirty bit

63. Which of the following algorithms is used to find the articulation points in a graph?
a) Bellman-Ford algorithm
b) Floyd-Warshall algorithm
c) Depth First Search
d) Kruskal's algorithm

64. In the N-Queens problem, what does backtracking do when a placed queen results in no valid positions for the next queen?
a) Stops the program
b) Recalculates all possible placements for the queens
c) Removes the last placed queen and tries the next possibility
d) Ignores the invalid position and moves forward

65. Which data structures are used to formulate the Breadth-first Search (BFS)?
a) Stack b) Queue c) Heap d) Linked List

66. Which strategy is used to handle deadlocks by ensuring that at least one of the necessary conditions for deadlock does not hold?
a) Deadlock Avoidance
b) Deadlock Detection and Recovery
c) Deadlock Prevention
d) Process Termination

67. Which disk scheduling algorithm minimizes the total seek time?
a) First-Come, First-Served (FCFS)
b) Shortest Seek Time First (SSTF)
c) Elevator (SCAN)
d) Circular SCAN (C-SCAN)

68. Access matrix model for user authentication contains ___________
a) A list of objects
b) A list of domains
c) A function which returns an object's type
d) All of these

69. A system digitizes an analog signal with a maximum frequency of 4 kHz. According to the Nyquist theorem, what is the minimum sampling rate required to avoid aliasing?
a) 4 kHz b) 8 kHz
c) 16 kHz d) 32 kHz

70. Using Go-Back-N ARQ with a window size of 4, how many frames can be sent before an acknowledgment is required?
a) 1 frame b) 4 frames c) 5 frames d) 8 frames

71. Which layer in the OSI model is responsible for data presentation?
a) Network layer b) Session layer
c) Presentation layer d) Transport layer

72. The purpose of object-oriented analysis is to_______
a) Identify the objects and their attributes
b) Define the system requirements
c) Model the system using object-oriented concepts
d) All of these

73. Use case diagram is used to __________
a) Model the interactions between users and the system
b) Model the classes and their relationships in the system
c) Model the flow of activities in the system
d) None of these

74. The purpose of the factory method design pattern is to _______

a) To define an interface for creating objects without specifying their concrete classes
b) To encapsulate object creation
c) To manage object destruction
d) To provide a way for multiple inheritance

75. Which of the following is TRUE?

a) Every relation in 3NF is also in BCNF
b) A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent on every key of R
c) Every relation in BCNF is also in 3NF
d) No relation can be in both BCNF and 3NF

76. The five items: A, B, C, D, and E are pushed in a stack, one after other starting from A. The stack is popped four times and each element is inserted in a queue. The two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is__________

a) A   b) B   c) C   d) D

77. If an array is declared as int arr[5][5], how many elements can it store?

a) 5   b) 25   c) 10   d) 0

78. Which of the binary tree traversal requires a Queue?

a) Level-order   b) pre-order
c) post-order   d) in-order

79. If R is a relation on set A= {1, 2, 3, 4} defined by R= {(x,y) | x+y=5}, then which pairs belong to R?

a) {(1,4),(2,3),(3,2),(4,1)}
b) {(1, 4), (2, 3), (4, 1)}
c) {(1, 4), (2, 3), (3, 2)}
d) {(1, 3), (2, 3), (4, 1)}

80. What is the Cartesian product of set A and set B, if the set A = {1, 2} and set B = {a, b}?

a) { (1, a), (1, b), (2, a), (b, b) }
b) { (1, 1), (2, 2), (a, a), (b, b) }
c) { (1, a), (2, a), (1, b), (2, b) }
d) { (1, 1), (a, a), (2, a), (1, b) }

81. A cycle on n vertices is isomorphic to its complement. What is the value of n?
a) 5   b) 8   c) 17   d) 32

82. Which of the following is a type of software testing that checks the functionality of the software in real-world scenarios?
a) Unit testing
b) Integration testing
c) System testing
d) User acceptance testing

83. Which one is the first phase of Rational Unified Process?
a) Elaboration b) Transition c) Construction d) Inception

84. Modifying a software after it has been put into use is known as
a) Software modification b) Software change
c) Software reengineering d) Software maintenance

85. If a system uses memory-mapped I/O, how much addressable memory is available if the address bus is 16 bits and 256 I/O addresses are reserved?
a) 65,536 bytes b) 65,280 bytes c) 32,768 bytes d) 32,512 bytes

86. Calculate the effective address for the instruction 'LOAD R1, 100(R2)' if the content of R2 is 200
a) 100 b) 200 c) 300 d) 400

87. Which of the following is an example of an I/O control method where the CPU polls each device?
a) Interrupt-driven I/O
b) Direct Memory Access (DMA)
c) Programmed I/O
d) Memory-mapped I/O

88. Suppose you have coins of denominations 1, 3 and 4. You use a greedy algorithm, in which you choose the largest denomination coin which is not greater than the remaining sum. For which of the following sums, will the algorithm NOT produce an optimal answer?
a) 20 b) 6 c) 12 d) 5

89. The worst case time complexity of quick sort is ______
a) O(n) b) $O(n^2)$ c) O(n log n) d) O(log n)

90. Which of the following algorithm is not in-place?
a) Selection sort b) Quick sort c) Merge Sort d) Bubble Sort

91. What is the main disadvantage of the **SSTF (Shortest Seek Time First)** disk scheduling algorithm?
a) It increases seek time compared to FCFS
b) It causes starvation for distant requests
c) It requires additional hardware support
d) It is less efficient than SCAN

92. Which of the following replacement policies suffers from Belady's anomaly?
a) LRU b) FIFO c) Optimal d) NRU

93. Thrashing occurs when :
a) The CPU is overloaded
b) Processes spend more time paging than executing
c) All processes are in a ready state
d) There is insufficient CPU scheduling

94. Which protocol type does Wi-Fi use for media access control?
a) FDMA b) CDMA c) CSMA/CA d) TDMA

95. In Ethernet addressing, if all the bits are 1s, the address is ____________
a) Unicast b) Multicast c) Broadcast d) None of these

96. Standard Ethernet uses_________ encoding
a) AMI b) Manchester
c) Differential Manchester d) NRZ

97. Which phase of the Unified Process involves deploying the system to its users?
a) Inception b) Elaboration
c) Construction d) Transition

98. In a weather monitoring system the **Observer** pattern is implemented. The system has the following components:

A **WeatherStation** object (subject) that notifies its observers whenever the temperature changes.

There are 200 **Display** objects (observers) that are subscribed to the **WeatherStation** to receive updates.

If each observer object stores the temperature (which is an integer) and has additional overhead of 4 bytes per observer for other attributes, calculate the total memory required to store all observer objects

a) 800 bytes b) 1200 bytes c) 1600 bytes d) 2000 bytes

99. _________________ Design pattern converts the interface of a class into another interface clients expect
a) Bridge method b) Adapter method
c) Proxy method d) Decorator method

100. Which of the following is a valid SQL aggregate function?
a) COUNT b) AVG c) MAX d) All of these.

* * * * *

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Question Paper Version : B

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Key Answers (CS/IS): **Version B**

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55	C	65	B	75	B	85	A	95	C
56	B	66	C	76	C	86	C	96	D
57	D	67	C	77	B	87	B	97	B
58	C	68	B	78	D	88	B	98	A
59	B	69	B	79	B	89	A	99	A
60	D	70	C	80	B	90	B	100	C

PART – I

(Research Methodology)

[50 Marks]

1. Which of the following best defines a random experiment?
a) A process that results in one of a set of outcomes, with certainty.
b) A process with uncertain outcomes that can be repeated under identical conditions.
c) A deterministic process where outcomes are predictable.
d) A set of outcomes that always yield the same result.

2. The probability of the occurrence of a single event, such as P(A) or P(B), which takes a specific value irrespective of the values of other events, i.e. unconditioned on any other events, is called?
a) Unconditional Probability b) Marginal Probability
c) Conditional Probability d) Joint Probability

3. What is the probability mass function (PMF) used to describe?
a) The likelihood of each outcome in a discrete probability distribution.
b) The cumulative probability of outcomes in a continuous distribution.
c) The variance of a continuous random variable.
d) The mean of a discrete random variable.

4. What does the Axiom of Probability P(S) = 1 mean, where S is the sample space?
a) The probability of any event is 0.
b) The probability of the sample space is 0.
c) The probability of the sample space is 1.
d) The probability of any event is 1.

5. Which of the following is true about a random variable?
a) It takes a fixed set of values.
b) It is a function that assigns a numerical value to each outcome in a sample space.
c) It has a deterministic relationship with other random variables.
d) It is not subject to probability distributions.

6. The binomial distribution is used to model which of the following?
a) The number of successes in a fixed number of independent trials with two possible outcomes.
b) The time between events in a fixed interval.
c) The number of occurrences of an event in a fixed interval of time.
d) The number of unique outcomes in a sample space.

7. The normal distribution is characterized by which of the following?
a) A skewed distribution with a peak at the right.
b) A symmetric bell-shaped curve where the mean, median, and mode are equal.
c) A discrete distribution with a fixed number of outcomes.
d) A distribution that models time between events.

8. Under which condition can the binomial distribution be approximated by a normal distribution?
a) When the number of trials is very small
b) When the probability of success is very high
c) When the probability of success is close to 0.5
d) When the trials are dependent

9. The Poisson distribution is typically used to model:
a) The number of successes in a fixed number of trials
b) The number of events occurring in a fixed interval of time or space
c) The time between events in a Poisson process
d) The number of distinct outcomes in a discrete sample

10. The hypergeometric distribution is used to model which of the following?
a) The number of successes in a fixed number of independent trials with two outcomes
b) The number of successes when the probability of success is constant across trials
c) The number of events occurring in a fixed interval of time
d) The number of successes in a fixed number of trials without replacement from a finite population

11. What does the analysis of variance (anova) primarily test for?
a) Differences in means across multiple groups
b) Differences in medians across two groups
c) Relationships between two variables
d) Equality of variances across groups

12. What is the main difference between one-way and two-way ANOVA?
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c) Independent variables depend on dependent variables.
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c) R-type factor analysis is applied to variables, whereas Q-type is applied to cases (individuals)
d) There is no significant difference between R-type and Q-type factor analysis

21. Type-I errors occurs if ________
a) Null Hypothesis is rejected even if it is true
b) Null Hypothesis is accepted even if it is false
c) Both Null hypothesis and alternative hypothesis are rejected
d) Both Null hypothesis and alternative hypothesis are accepted

22. Which is a preferred sampling method for the population of a finite size.
a) Systematic sampling
b) Purposive Sampling
c) Cluster Sampling
d) Area Sampling

23. Which of the following is a probability sampling method?
a) Convenience sampling
b) Stratified random sampling
c) Quota sampling
d) Judgmental sampling

24. Which classification of measurement scale represents ordered categories but does not have a true zero point?
a) Nominal scale b) Ordinal scale
c) Interval scale d) Ratio scale

25. Which method is most suitable for collecting primary data when studying behaviour over time?
a) Case study method
b) Experimental method
c) Longitudinal survey
d) Secondary data collection

26. Which of the following is NOT a step in the data preparation process?
a) Editing
b) Coding
c) Data collection
d) Transcribing

27. What type of analysis focuses on summarizing the data collected?
a) Inferential analysis
b) Diagnostic analysis
c) Predictive analysis
d) Descriptive analysis

28. Which measure of central tendency is best suited for categorical data?
a) Mean
b) Median
c) Mode
d) Standard deviation

29. What does skewness measure in a dataset?
a) The average value of the data
b) The spread of data points
c) The relationship between variables
d) The symmetry of the data distribution

30. What is the correlation coefficient used to measure?
a) The degree of dispersion in the data
b) The relationship between two variables
c) The central value of a dataset
d) The extent of symmetry in the distribution

31. What is the main objective of applied research?
a) Develop theories for future studies
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d) To summarize the results

37. Which of the following is NOT a feature of a good research design?
a) Flexibility
b) Objectivity
c) Economical execution
d) Bias in data collection

38. Which of the following is a principle of experimental design?
a) Randomization b) Subjectivity
c) Hypothesis elimination d) Literature review

39. What is a conceptual framework?
a) A plan for data analysis
b) A framework to explain concepts and relationships in the research
c) A compilation of all references
d) A step-by-step experimental protocol

40. Which type of research design is best for studying cause-and-effect relationships?
a) Descriptive design
b) Correlational design
c) Experimental design
d) Exploratory design

41. Which of the following is an example of a non-sampling error?
a) Errors in calculating the standard error
b) Errors due to small sample size
c) Errors due to incomplete responses
d) Errors caused by random sampling

42. What does the Central Limit Theorem state?
a) The sampling distribution of the mean approaches a normal distribution as sample size increases
b) The mean of the population equals the mean of the sample
c) The variance of the population is less than that of the sample
d) The sampling method used always determines the distribution

43. What is the standard error?
a) The difference between the sample mean and population mean
b) The standard deviation of the sampling distribution
c) The sum of all errors in data collection
d) The variance of a data set

44. What is the main goal of statistical inference?
a) To describe the collected data
b) To evaluate the quality of the data
c) To calculate the mean of the sample
d) To make predictions about the population based on the sample

45. What is a sampling distribution?
a) The distribution of the population data
b) The histogram of the sampled data
c) The error associated with a specific sample
d) The distribution of a statistic derived from all possible samples of the same size

46. What does the P-value represent in hypothesis testing?
a) The probability of making a Type II error
b) The probability of observing the data if the null hypothesis is true
c) The critical value for the test statistic
d) The standard deviation of the sample

47. Which test is used to compare the means of two independent samples?
a) Chi-square test
b) Z-test for proportion
c) Two-sample t-test
d) ANOVA

48. What does the chi-square goodness-of-fit test evaluate?
a) The mean difference between two groups
b) Whether observed data follows a specific distribution
c) The equality of variances between two samples
d) The correlation between two variables

49. Which hypothesis test is used to compare the variances of two populations?
a) Z-test for variance
b) F-test
c) Paired t-test
d) Chi-square test for independence

50 What is a null hypothesis (H_0)?
a) A hypothesis that predicts a significant relationship between variables
b) A hypothesis that states there is no significant effect or relationship
c) A hypothesis tested only for large samples
d) A hypothesis that must always be true

* * * * *

PART – II
Discipline Oriented Section
(Computer Science and Engineering / Information Science and Engineering)

[50 MARKS]

51. What is the main disadvantage of the **SSTF (Shortest Seek Time First)** disk scheduling algorithm?
a) It increases seek time compared to FCFS
b) It causes starvation for distant requests
c) It requires additional hardware support
d) It is less efficient than SCAN

52. Which of the following replacement policies suffers from Belady's anomaly?
a) LRU b) FIFO c) Optimal d) NRU

53. Thrashing occurs when :
a) The CPU is overloaded
b) Processes spend more time paging than executing
c) All processes are in a ready state
d) There is insufficient CPU scheduling

54. Which protocol type does Wi-Fi use for media access control?
a) FDMA b) CDMA c) CSMA/CA d) TDMA

55. In Ethernet addressing, if all the bits are 1s, the address is ____________
a) Unicast b) Multicast c) Broadcast d) None of these

56. Standard Ethernet uses_________ encoding
a) AMI b) Manchester
c) Differential Manchester d) NRZ

57. Which phase of the Unified Process involves deploying the system to its users?
a) Inception b) Elaboration
c) Construction d) Transition

58. In a weather monitoring system the **Observer** pattern is implemented. The system has the following components:

A **WeatherStation** object (subject) that notifies its observers whenever the temperature changes.

There are 200 **Display** objects (observers) that are subscribed to the **WeatherStation** to receive updates.

If each observer object stores the temperature (which is an integer) and has additional overhead of 4 bytes per observer for other attributes, calculate the total memory required to store all observer objects
a) 800 bytes b) 1200 bytes c) 1600 bytes d) 2000 bytes

59. ________________ Design pattern converts the interface of a class into another interface clients expect

a) Bridge method
b) Adapter method
c) Proxy method
d) Decorator method

60. Which of the following is a valid SQL aggregate function?

a) COUNT
b) AVG
c) MAX
d) All of these.

61. A cycle on n vertices is isomorphic to its complement. What is the value of n?

a) 5
b) 8
c) 17
d) 32

62. Which of the following is a type of software testing that checks the functionality of the software in real-world scenarios?

a) Unit testing
b) Integration testing
c) System testing
d) User acceptance testing

63. Which one is the first phase of Rational Unified Process?

a) Elaboration
b) Transition
c) Construction
d) Inception

64. Modifying a software after it has been put into use is known as

a) Software modification
b) Software change
c) Software reengineering
d) Software maintenance

65. If a system uses memory-mapped I/O, how much addressable memory is available if the address bus is 16 bits and 256 I/O addresses are reserved?

a) 65,536 bytes
b) 65,280 bytes
c) 32,768 bytes
d) 32,512 bytes

66. Calculate the effective address for the instruction 'LOAD R1, 100(R2)' if the content of R2 is 200

a) 100
b) 200
c) 300
d) 400

67. Which of the following is an example of an I/O control method where the CPU polls each device?

a) Interrupt-driven I/O
b) Direct Memory Access (DMA)
c) Programmed I/O
d) Memory-mapped I/O

68. Suppose you have coins of denominations 1, 3 and 4. You use a greedy algorithm, in which you choose the largest denomination coin which is not greater than the remaining sum. For which of the following sums, will the algorithm NOT produce an optimal answer?

a) 20 b) 6
c) 12 d) 5

69. The worst case time complexity of quick sort is ______

a) O(n) b) $O(n^2)$
c) O(n log n) d) O(log n)

70. Which of the following algorithm is not in-place?

a) Selection sort b) Quick sort
c) Merge Sort d) Bubble Sort

71. The unit which acts as an intermediate agent between memory and backing store to reduce process time is ___________

a) Translation Lookaside Buffer's
b) Registers
c) Page Tables
d) Cache

72. The bit used to signify that the cache location is updated is ________

a) Flag bit b) Reference bit
c) Update bit d) Dirty bit

73. Which of the following algorithms is used to find the articulation points in a graph?

a) Bellman-Ford algorithm
b) Floyd-Warshall algorithm
c) Depth First Search
d) Kruskal's algorithm

74. In the N-Queens problem, what does backtracking do when a placed queen results in no valid positions for the next queen?

a) Stops the program
b) Recalculates all possible placements for the queens
c) Removes the last placed queen and tries the next possibility
d) Ignores the invalid position and moves forward

75. Which data structures are used to formulate the Breadth-first Search (BFS)?

a) Stack b) Queue
c) Heap d) Linked List

76. Which strategy is used to handle deadlocks by ensuring that at least one of the necessary conditions for deadlock does not hold?
a) Deadlock Avoidance
b) Deadlock Detection and Recovery
c) Deadlock Prevention
d) Process Termination

77. Which disk scheduling algorithm minimizes the total seek time?
a) First-Come, First-Served (FCFS)
b) Shortest Seek Time First (SSTF)
c) Elevator (SCAN)
d) Circular SCAN (C-SCAN)

78. Access matrix model for user authentication contains ___________
a) A list of objects b) A list of domains
c) A function which returns an object's type d) All of these

79. A system digitizes an analog signal with a maximum frequency of 4 kHz. According to the Nyquist theorem, what is the minimum sampling rate required to avoid aliasing?
a) 4 kHz b) 8 kHz c) 16 kHz d) 32 kHz

80. Using Go-Back-N ARQ with a window size of 4, how many frames can be sent before an acknowledgment is required?
a) 1 frame b) 4 frames c) 5 frames d) 8 frames

81. Predict the output of following program

```
#include <stdio.h>
int fun(int n)
{
   if (n == 4)
      return n;
   else return 2*fun(n+1);
}
int main()
{
  printf("%d", fun(2));
  return 0;
}
```

a) 4 b) 8 c) 16 d) Runtime Error

82. A Binary search tree is generated by inserting the following elements in order,
50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24
The number of nodes in the left sub tree and right sub tree of the root is ________
a) (4, 7) b) (7, 4) c) (8, 3) d) (3, 8)

83. A variant of the linked list in which none of the node contains NULL pointer is______
a) Singly linked list b) Doubly linked list
c) Circular linked list d) None

84. If p and q are independent propositions, which of these is always false?
a) $p \wedge \neg q$ b) $\neg p \wedge \neg q$ c) $p \rightarrow q$ d) $p \leftrightarrow \neg q$

85. How many equivalence relations can be defined on a set of size 3?
a) 5 b) 10 c) 15 d) 6

86. In a 7-node directed cyclic graph, the number of Hamiltonian cycle is to be
a) 180 b) 720 c) 360 d) 540

87. What is the purpose of a Use Case Diagram in UML?
a) To model the flow of data within the system
b) To show the system's interactions with external entities
c) To define the class structure of the system
d) To represent the system's behavior over time

88. In the Rational Unified Process (RUP), which phase focuses on creating the architectural foundation of the system?
a) Inception b) Elaboration
c) Construction d) Transition

89. Which type of testing ensures that previously developed and tested software still works after changes?
a) Regression Testing b) Acceptance Testing
c) Integration Testing d) System Testing

90. Which of the following acts as a temporary storage location to hold an intermediate results for mathematical and logical calculations
a) RAM b) Accumulator
c) Program counter d) Memory Address Register

91. Which layer in the OSI model is responsible for data presentation?
a) Network layer b) Session layer c) Presentation layer d) Transport layer

92. The purpose of object-oriented analysis is to_______
a) Identify the objects and their attributes
b) Define the system requirements
c) Model the system using object-oriented concepts
d) All of these

93. Use case diagram is used to __________
a) Model the interactions between users and the system
b) Model the classes and their relationships in the system
c) Model the flow of activities in the system
d) None of these

94. The purpose of the factory method design pattern is to _______
a) To define an interface for creating objects without specifying their concrete classes
b) To encapsulate object creation
c) To manage object destruction
d) To provide a way for multiple inheritance

95. Which of the following is TRUE?
a) Every relation in 3NF is also in BCNF
b) A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent on every key of R
c) Every relation in BCNF is also in 3NF
d) No relation can be in both BCNF and 3NF

96. The five items: A, B, C, D, and E are pushed in a stack, one after other starting from A. The stack is popped four times and each element is inserted in a queue. The two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is__________
a) A b) B c) C d) D

97. If an array is declared as int arr[5][5], how many elements can it store?
a) 5 b) 25 c) 10 d) 0

98. Which of the binary tree traversal requires a Queue?
a) Level-order b) pre-order c) post-order d) in-order

99. If R is a relation on set A= {1, 2, 3, 4} defined by R= {(x,y) | x+y=5}, then which pairs belong to R?
a) {(1,4),(2,3),(3,2),(4,1)}
b) {(1, 4), (2, 3), (4, 1)}
c) {(1, 4), (2, 3), (3, 2)}
d) {(1, 3), (2, 3), (4, 1)}

100. What is the Cartesian product of set A and set B, if the set A = {1, 2} and set B = {a, b}?
a) { (1, a), (1, b), (2, a), (b, b) }
b) { (1, 1), (2, 2), (a, a), (b, b) }
c) { (1, a), (2, a), (1, b), (2, b) }
d) { (1, 1), (a, a), (2, a), (1, b) }

* * * * *

VTU-ETR Seat No. | | | | | | |

Question Paper Version : C

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research), January – 2025

DO NOT USE MOBILE PHONES

Computer Science and Engineering / Information Science and Engineering

Time: 3 hrs. Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
5. Missing data, if any, may be suitably assumed.
6. There shall be no negative marking for wrong answers.
7. The examinees shall select the response which they want to mark/indicate on the response sheet and shade/darken/blacken completely the inside area of the corresponding checkbox bubble of circular or oval shape.
8. Candidates should use black ball point pen to shade the checkbox bubble so that it is fully dark and the computer can detect without fail. Shading should be limited to the checkbox bubble only.
9. Shading of multiple checkbox bubbles shall be treated as invalid answer.
10. Partial or incomplete shading/darkening of the bubble be avoided.
11. Use of whiteners/correction fluid to change the shaded/filled checkbox bubble to fill another is prohibited.
12. Erasing of filled checkbox bubble/s to fill another is prohibited.
13. Any type of writing, scribbling etc., on the question paper and ticking the response of a question/s are prohibited. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
14. Only non-programmable scientific calculator is permitted. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
15. Folding/crushing/mutilating of OMR sheet is not permitted.
16. Handover the OMR sheet to the Room Invigilator while leaving the examination hall.

Key Answers (CS/IS): **Version C**

PART - I

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
1	a	11	b	21	b	31	c	41	a
2	b	12	c	22	c	32	a	42	a
3	b	13	b	23	a	33	b	43	b
4	a	14	c	24	c	34	d	44	c
5	c	15	c	25	b	35	d	45	c
6	a	16	b	26	a	36	b	46	c
7	d	17	d	27	b	37	c	47	d
8	d	18	a	28	c	38	b	48	c
9	b	19	b	29	b	39	b	49	d
10	c	20	c	30	b	40	b	50	b

PART – II

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
51	C	61	B	71	A	81	D	91	C
52	D	62	B	72	D	82	D	92	B
53	A	63	B	73	D	83	C	93	C
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c) Economical execution
d) Bias in data collection

18. Which of the following is a principle of experimental design?
a) Randomization
b) Subjectivity
c) Hypothesis elimination
d) Literature review

19. What is a conceptual framework?
a) A plan for data analysis
b) A framework to explain concepts and relationships in the research
c) A compilation of all references
d) A step-by-step experimental protocol

20. Which type of research design is best for studying cause-and-effect relationships?
a) Descriptive design
b) Correlational design
c) Experimental design
d) Exploratory design

21. Which of the following best defines a random experiment?
a) A process that results in one of a set of outcomes, with certainty.
b) A process with uncertain outcomes that can be repeated under identical conditions.
c) A deterministic process where outcomes are predictable.
d) A set of outcomes that always yield the same result.

22. The probability of the occurrence of a single event, such as P(A) or P(B), which takes a specific value irrespective of the values of other events, i.e. unconditioned on any other events, is called?
a) Unconditional Probability
b) Marginal Probability
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d) Joint Probability

23. What is the probability mass function (PMF) used to describe?
a) The likelihood of each outcome in a discrete probability distribution.
b) The cumulative probability of outcomes in a continuous distribution.
c) The variance of a continuous random variable.
d) The mean of a discrete random variable.

24. What does the Axiom of Probability P(S) = 1 mean, where S is the sample space?
a) The probability of any event is 0.
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25. Which of the following is true about a random variable?
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c) It has a deterministic relationship with other random variables.
d) It is not subject to probability distributions.

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a) A skewed distribution with a peak at the right.
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c) A discrete distribution with a fixed number of outcomes.
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28. Under which condition can the binomial distribution be approximated by a normal distribution?
a) When the number of trials is very small
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29. The Poisson distribution is typically used to model:
a) The number of successes in a fixed number of trials
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30. The hypergeometric distribution is used to model which of the following?
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a) Errors in calculating the standard error
b) Errors due to small sample size
c) Errors due to incomplete responses
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32. What does the Central Limit Theorem state?
a) The sampling distribution of the mean approaches a normal distribution as sample size increases
b) The mean of the population equals the mean of the sample
c) The variance of the population is less than that of the sample
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33. What is the standard error?
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c) The sum of all errors in data collection
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37. Which test is used to compare the means of two independent samples?
a) Chi-square test
b) Z-test for proportion
c) Two-sample t-test
d) ANOVA

38. What does the chi-square goodness-of-fit test evaluate?
a) The mean difference between two groups
b) Whether observed data follows a specific distribution
c) The equality of variances between two samples
d) The correlation between two variables

39. Which hypothesis test is used to compare the variances of two populations?
a) Z-test for variance
b) F-test
c) Paired t-test
d) Chi-square test for independence

40. What is a null hypothesis (H_0)?
a) A hypothesis that predicts a significant relationship between variables
b) A hypothesis that states there is no significant effect or relationship
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41. Type-I errors occurs if ________
a) Null Hypothesis is rejected even if it is true
b) Null Hypothesis is accepted even if it is false
c) Both Null hypothesis and alternative hypothesis are rejected
d) Both Null hypothesis and alternative hypothesis are accepted

42. Which is a preferred sampling method for the population of a finite size.
a) Systematic sampling
b) Purposive Sampling
c) Cluster Sampling
d) Area Sampling

43. Which of the following is a probability sampling method?
a) Convenience sampling
b) Stratified random sampling
c) Quota sampling
d) Judgmental sampling

44. Which classification of measurement scale represents ordered categories but does not have a true zero point?
a) Nominal scale b) Ordinal scale
c) Interval scale d) Ratio scale

45. Which method is most suitable for collecting primary data when studying behaviour over time?
a) Case study method
b) Experimental method
c) Longitudinal survey
d) Secondary data collection

46. Which of the following is NOT a step in the data preparation process?
a) Editing b) Coding c) Data collection d) Transcribing

47. What type of analysis focuses on summarizing the data collected?
a) Inferential analysis
b) Diagnostic analysis
c) Predictive analysis
d) Descriptive analysis

48. Which measure of central tendency is best suited for categorical data?
a) Mean b) Median
c) Mode d) Standard deviation

49. What does skewness measure in a dataset?
a) The average value of the data
b) The spread of data points
c) The relationship between variables
d) The symmetry of the data distribution

50 What is the correlation coefficient used to measure?
a) The degree of dispersion in the data
b) The relationship between two variables
c) The central value of a dataset
d) The extent of symmetry in the distribution

* * * * *

PART – II
Discipline Oriented Section
(Computer Science and Engineering / Information Science and Engineering)

[50 MARKS]

51. Which layer in the OSI model is responsible for data presentation?
a) Network layer b) Session layer
c) Presentation layer d) Transport layer

52. The purpose of object-oriented analysis is to_______
a) Identify the objects and their attributes
b) Define the system requirements
c) Model the system using object-oriented concepts
d) All of these

53. Use case diagram is used to __________
a) Model the interactions between users and the system
b) Model the classes and their relationships in the system
c) Model the flow of activities in the system
d) None of these

54. The purpose of the factory method design pattern is to _______
a) To define an interface for creating objects without specifying their concrete classes
b) To encapsulate object creation
c) To manage object destruction
d) To provide a way for multiple inheritance

55. Which of the following is TRUE?
a) Every relation in 3NF is also in BCNF
b) A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent on every key of R
c) Every relation in BCNF is also in 3NF
d) No relation can be in both BCNF and 3NF

56. The five items: A, B, C, D, and E are pushed in a stack, one after other starting from A. The stack is popped four times and each element is inserted in a queue. The two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is__________
a) A b) B c) C d) D

57. If an array is declared as int arr[5][5], how many elements can it store?
a) 5 b) 25 c) 10 d) 0

58. Which of the binary tree traversal requires a Queue?
a) Level-order b) pre-order c) post-order d) in-order

59. If R is a relation on set A= {1, 2, 3, 4} defined by R= {(x,y) | x+y=5}, then which pairs belong to R?
a) {(1,4),(2,3),(3,2),(4,1)}
b) {(1, 4), (2, 3), (4, 1)}
c) {(1, 4), (2, 3), (3, 2)}
d) {(1, 3), (2, 3), (4, 1)}

60. What is the Cartesian product of set A and set B, if the set A = {1, 2} and set B = {a, b}?
a) { (1, a), (1, b), (2, a), (b, b) }
b) { (1, 1), (2, 2), (a, a), (b, b) }
c) { (1, a), (2, a), (1, b), (2, b) }
d) { (1, 1), (a, a), (2, a), (1, b) }

61. What is the main disadvantage of the **SSTF (Shortest Seek Time First)** disk scheduling algorithm?
a) It increases seek time compared to FCFS
b) It causes starvation for distant requests
c) It requires additional hardware support
d) It is less efficient than SCAN

62. Which of the following replacement policies suffers from Belady's anomaly?
a) LRU b) FIFO c) Optimal d) NRU

63. Thrashing occurs when :
a) The CPU is overloaded
b) Processes spend more time paging than executing
c) All processes are in a ready state
d) There is insufficient CPU scheduling

64. Which protocol type does Wi-Fi use for media access control?
a) FDMA b) CDMA c) CSMA/CA d) TDMA

65. In Ethernet addressing, if all the bits are 1s, the address is ____________
a) Unicast b) Multicast c) Broadcast d) None of these

66. Standard Ethernet uses_________ encoding
a) AMI b) Manchester c) Differential Manchester d) NRZ

67. Which phase of the Unified Process involves deploying the system to its users?
a) Inception b) Elaboration c) Construction d) Transition

68. In a weather monitoring system the **Observer** pattern is implemented. The system has the following components:

A **WeatherStation** object (subject) that notifies its observers whenever the temperature changes.

There are 200 **Display** objects (observers) that are subscribed to the **WeatherStation** to receive updates.

If each observer object stores the temperature (which is an integer) and has additional overhead of 4 bytes per observer for other attributes, calculate the total memory required to store all observer objects

a) 800 bytes b) 1200 bytes c) 1600 bytes d) 2000 bytes

69. ________________ Design pattern converts the interface of a class into another interface clients expect

a) Bridge method b) Adapter method c) Proxy method d) Decorator method

70. Which of the following is a valid SQL aggregate function?

a) COUNT b) AVG c) MAX d) All of these.

71. A cycle on n vertices is isomorphic to its complement. What is the value of n?

a) 5 b) 8 c) 17 d) 32

72. Which of the following is a type of software testing that checks the functionality of the software in real-world scenarios?

a) Unit testing
b) Integration testing
c) System testing
d) User acceptance testing

73. Which one is the first phase of Rational Unified Process?

a) Elaboration b) Transition c) Construction d) Inception

74. Modifying a software after it has been put into use is known as

a) Software modification
b) Software change
c) Software reengineering
d) Software maintenance

75. If a system uses memory-mapped I/O, how much addressable memory is available if the address bus is 16 bits and 256 I/O addresses are reserved?

a) 65,536 bytes b) 65,280 bytes c) 32,768 bytes d) 32,512 bytes

76. Calculate the effective address for the instruction 'LOAD R1, 100(R2)' if the content of R2 is 200

a) 100 b) 200 c) 300 d) 400

77. Which of the following is an example of an I/O control method where the CPU polls each device?
a) Interrupt-driven I/O
b) Direct Memory Access (DMA)
c) Programmed I/O
d) Memory-mapped I/O

78. Suppose you have coins of denominations 1, 3 and 4. You use a greedy algorithm, in which you choose the largest denomination coin which is not greater than the remaining sum. For which of the following sums, will the algorithm NOT produce an optimal answer?
a) 20 b) 6 c) 12 d) 5

79. The worst case time complexity of quick sort is ______
a) O(n) b) $O(n^2)$ c) O(n log n) d) O(log n)

80. Which of the following algorithm is not in-place?
a) Selection sort b) Quick sort c) Merge Sort d) Bubble Sort

81. The unit which acts as an intermediate agent between memory and backing store to reduce process time is ___________
a) Translation Lookaside Buffer's
b) Registers
c) Page Tables
d) Cache

82. The bit used to signify that the cache location is updated is ________
a) Flag bit b) Reference bit
c) Update bit d) Dirty bit

83. Which of the following algorithms is used to find the articulation points in a graph?
a) Bellman-Ford algorithm
b) Floyd-Warshall algorithm
c) Depth First Search
d) Kruskal's algorithm

84. In the N-Queens problem, what does backtracking do when a placed queen results in no valid positions for the next queen?
a) Stops the program
b) Recalculates all possible placements for the queens
c) Removes the last placed queen and tries the next possibility
d) Ignores the invalid position and moves forward

85. Which data structures are used to formulate the Breadth-first Search (BFS)?
a) Stack b) Queue c) Heap d) Linked List

86. Which strategy is used to handle deadlocks by ensuring that at least one of the necessary conditions for deadlock does not hold?
a) Deadlock Avoidance
b) Deadlock Detection and Recovery
c) Deadlock Prevention
d) Process Termination

87. Which disk scheduling algorithm minimizes the total seek time?
a) First-Come, First-Served (FCFS)
b) Shortest Seek Time First (SSTF)
c) Elevator (SCAN)
d) Circular SCAN (C-SCAN)

88. Access matrix model for user authentication contains ___________
a) A list of objects
b) A list of domains
c) A function which returns an object's type
d) All of these

89. A system digitizes an analog signal with a maximum frequency of 4 kHz. According to the Nyquist theorem, what is the minimum sampling rate required to avoid aliasing?
a) 4 kHz b) 8 kHz c) 16 kHz d) 32 kHz

90. Using Go-Back-N ARQ with a window size of 4, how many frames can be sent before an acknowledgment is required?
a) 1 frame b) 4 frames c) 5 frames d) 8 frames

91. Predict the output of following program

```
#include <stdio.h>
int fun(int n)
{
   if (n == 4)
      return n;
   else return 2*fun(n+1);
}
int main()
{
  printf("%d", fun(2));
  return 0;
}
```

a) 4 b) 8 c) 16 d) Runtime Error

92. A Binary search tree is generated by inserting the following elements in order,
50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24
The number of nodes in the left sub tree and right sub tree of the root is ________
a) (4, 7) b) (7, 4) c) (8, 3) d) (3, 8)

93. A variant of the linked list in which none of the node contains NULL pointer is______

a) Singly linked list b) Doubly linked list
c) Circular linked list d) None

94. If p and q are independent propositions, which of these is always false?

a) $p \wedge \neg q$ b) $\neg p \wedge \neg q$ c) $p \rightarrow q$ d) $p \leftrightarrow \neg q$

95. How many equivalence relations can be defined on a set of size 3?

a) 5 b) 10 c) 15 d) 6

96. In a 7-node directed cyclic graph, the number of Hamiltonian cycle is to be

a) 180 b) 720 c) 360 d) 540

97. What is the purpose of a Use Case Diagram in UML?

a) To model the flow of data within the system
b) To show the system's interactions with external entities
c) To define the class structure of the system
d) To represent the system's behavior over time

98. In the Rational Unified Process (RUP), which phase focuses on creating the architectural foundation of the system?

a) Inception b) Elaboration c) Construction d) Transition

99. Which type of testing ensures that previously developed and tested software still works after changes?

a) Regression Testing b) Acceptance Testing
c) Integration Testing d) System Testing

100. Which of the following acts as a temporary storage location to hold an intermediate results for mathematical and logical calculations

a) RAM
b) Accumulator
c) Program counter
d) Memory Address Register

* * * * *

VTU-ETR Seat No. | | | | | | |

Question Paper Version : D

Visvesvaraya Technological University, Belagavi

Eligibility Test for Research (VTU-ETR) Programmes

Ph.D./M.S. (Research), January – 2025

Computer Science and Engineering / Information Science and Engineering

Time: 3 hrs. Max. Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. Ensure that the question paper booklet issued to you is of your opted discipline.
2. If your name, ETR number, OMR No., branch and branch code are not already printed on the OMR answer sheet, write them in clear handwriting in the space provided.
3. The question paper has 100 multiple choice questions. Each question carries one mark.
4. All the questions are compulsory.
5. Missing data, if any, may be suitably assumed.
6. There shall be no negative marking for wrong answers.
7. The examinees shall select the response which they want to mark/indicate on the response sheet and shade/darken/blacken completely the inside area of the corresponding checkbox bubble of circular or oval shape.
8. Candidates should use black ball point pen to shade the checkbox bubble so that it is fully dark and the computer can detect without fail. Shading should be limited to the checkbox bubble only.
9. Shading of multiple checkbox bubbles shall be treated as invalid answer.
10. Partial or incomplete shading/darkening of the bubble be avoided.
11. Use of whiteners/correction fluid to change the shaded/filled checkbox bubble to fill another is prohibited.
12. Erasing of filled checkbox bubble/s to fill another is prohibited.
13. Any type of writing, scribbling etc., on the question paper and ticking the response of a question/s are prohibited. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
14. Only non-programmable scientific calculator is permitted. Violation of this condition shall lead to disciplinary action like debar from the examination. Also, the OMR answer sheet shall not be considered for evaluation process.
15. Folding/crushing/mutilating of OMR sheet is not permitted.
16. Handover the OMR sheet to the Room Invigilator while leaving the examination hall.

Key Answers (CS/IS): **Version D**

PART - I

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
1	**c**	11	**b**	21	**a**	31	**a**	41	**b**
2	**a**	12	**c**	22	**b**	32	**a**	42	**c**
3	**b**	13	**a**	23	**b**	33	**b**	43	**b**
4	**d**	14	**c**	24	**a**	34	**c**	44	**c**
5	**d**	15	**b**	25	**c**	35	**c**	45	**c**
6	**b**	16	**a**	26	**a**	36	**c**	46	**b**
7	**c**	17	**b**	27	**d**	37	**d**	47	**d**
8	**b**	18	**c**	28	**d**	38	**c**	48	**a**
9	**b**	19	**b**	29	**b**	39	**d**	49	**b**
10	**b**	20	**b**	30	**c**	40	**b**	50	**c**

PART – II

Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer	Q.No	Answer
51	A	61	C	71	B	81	C	91	D
52	D	62	B	72	B	82	D	92	D
53	D	63	C	73	B	83	A	93	C
54	D	64	D	74	C	84	A	94	C
55	B	65	A	75	C	85	C	95	B
56	C	66	C	76	B	86	D	96	C
57	C	67	B	77	D	87	B	97	B
58	B	68	B	78	C	88	A	98	D
59	B	69	A	79	B	89	A	99	B
60	C	70	B	80	D	90	C	100	B

PART – I

(Research Methodology)

[50 Marks]

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   c) Two-sample t-test
   d) ANOVA

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c) The number of events occurring in a fixed interval of time
d) The number of successes in a fixed number of trials without replacement from a finite population

21. What does the analysis of variance (anova) primarily test for?
a) Differences in means across multiple groups
b) Differences in medians across two groups
c) Relationships between two variables
d) Equality of variances across groups

22. What is the main difference between one-way and two-way ANOVA?
a) One-way ANOVA tests one group, while two-way ANOVA tests two groups
b) One-way ANOVA examines one independent variable, while two-way ANOVA examines two independent variables
c) Two-way ANOVA is only for experimental designs
d) One-way ANOVA includes interaction effects

23. In a Latin-square design, how are the treatments arranged?
a) Randomly across all rows
b) Such that each treatment appears once in each row and column
c) To maximize variability in the design
d) Only within rows and not columns

24. What is the F-ratio in ANOVA?
a) The ratio of between-group variance to within-group variance
b) The ratio of group means to total variance
c) The sum of squares within groups
d) The mean of all variances

25. Which of the following is NOT an assumption of ANOVA?
a) Homogeneity of variances
b) Independence of observations
c) Equal sample sizes in all groups
d) Normality of residuals

26. Which of the following is a key characteristic of dependent and independent variables in linear regression?
a) Dependent variable is predicted by the independent variable.
b) Both dependent and independent variables are random variables.
c) Independent variables depend on dependent variables.
d) Dependent variables are not affected by independent variables.

27. What problem arises when independent variables in a multiple regression model are highly correlated with each other?
a) Heteroscedasticity
b) Homoscedasticity
c) Autocorrelation
d) Multicollinearity

28. Which of the following is an example of a qualitative explanatory variable in regression analysis?
a) Age of a person
b) Height of an individual
c) Weight of an individual
d) Type of education (e.g., High School, College, Graduate)

29. What is the primary purpose of factor analysis?
a) To predict future values based on historical data.
b) To reduce the number of variables by identifying underlying factors.
c) To determine the causal relationship between variables.
d) To test hypotheses in a regression model

30. Which of the following is a key difference between R-type and Q-type factor analysis?
a) R-type focuses on factors affecting the variables, while Q-type analyzes the variables' impact on the factors
b) R-type is used when analyzing observations across multiple time periods, while Q-type focuses on cross-sectional data
c) R-type factor analysis is applied to variables, whereas Q-type is applied to cases (individuals)
d) There is no significant difference between R-type and Q-type factor analysis

31. Type-I errors occurs if ________
a) Null Hypothesis is rejected even if it is true
b) Null Hypothesis is accepted even if it is false
c) Both Null hypothesis and alternative hypothesis are rejected
d) Both Null hypothesis and alternative hypothesis are accepted

32. Which is a preferred sampling method for the population of a finite size.
a) Systematic sampling
b) Purposive Sampling
c) Cluster Sampling
d) Area Sampling

33. Which of the following is a probability sampling method?
a) Convenience sampling
b) Stratified random sampling
c) Quota sampling
d) Judgmental sampling

34. Which classification of measurement scale represents ordered categories but does not have a true zero point?
a) Nominal scale b) Ordinal scale
c) Interval scale d) Ratio scale

35. Which method is most suitable for collecting primary data when studying behaviour over time?
a) Case study method
b) Experimental method
c) Longitudinal survey
d) Secondary data collection

36. Which of the following is NOT a step in the data preparation process?
a) Editing b) Coding c) Data collection d) Transcribing

37. What type of analysis focuses on summarizing the data collected?
a) Inferential analysis
b) Diagnostic analysis
c) Predictive analysis
d) Descriptive analysis

38. Which measure of central tendency is best suited for categorical data?
a) Mean b) Median c) Mode d) Standard deviation

39. What does skewness measure in a dataset?
a) The average value of the data
b) The spread of data points
c) The relationship between variables
d) The symmetry of the data distribution

40. What is the correlation coefficient used to measure?
a) The degree of dispersion in the data
b) The relationship between two variables
c) The central value of a dataset
d) The extent of symmetry in the distribution

41. What is the main objective of applied research?
a) Develop theories for future studies
b) Solve specific, practical problems
c) Investigate unexplored phenomena
d) Study the historical context of an issue

42. Which of the following steps is NOT part of the research process?
a) Defining the research problem
b) Reviewing related literature
c) Avoiding data collection
d) Formulating a hypothesis

43. Which of the following is NOT a characteristic of good research?
a) Systematic and logical
b) Based on subjective interpretation
c) Replicable and verifiable
d) Directed towards problem-solving

44. Why is it necessary to define a research problem clearly?
a) To reduce the scope of the study
b) To formulate irrelevant hypotheses
c) To guide the research process effectively
d) To avoid reviewing related literature

45. Which research approach combines both qualitative and quantitative methods?
a) Descriptive research
b) Experimental research
c) Mixed-method research
d) Longitudinal research

46. What is the primary purpose of a literature review in research?
a) To test hypotheses
b) To identify gaps in existing knowledge
c) To collect primary data
d) To summarize the results

47. Which of the following is NOT a feature of a good research design?
a) Flexibility
b) Objectivity
c) Economical execution
d) Bias in data collection

48. Which of the following is a principle of experimental design?
a) Randomization
b) Subjectivity
c) Hypothesis elimination
d) Literature review

49. What is a conceptual framework?
a) A plan for data analysis
b) A framework to explain concepts and relationships in the research
c) A compilation of all references
d) A step-by-step experimental protocol

50 Which type of research design is best for studying cause-and-effect relationships?
a) Descriptive design
b) Correlational design
c) Experimental design
d) Exploratory design

* * * * *

PART – II
Discipline Oriented Section
(Computer Science and Engineering / Information Science and Engineering)

[50 MARKS]

51. A cycle on n vertices is isomorphic to its complement. What is the value of n?
a) 5 b) 8 c) 17 d) 32

52. Which of the following is a type of software testing that checks the functionality of the software in real-world scenarios?
a) Unit testing
b) Integration testing
c) System testing
d) User acceptance testing

53. Which one is the first phase of Rational Unified Process?
a) Elaboration b) Transition
c) Construction d) Inception

54. Modifying a software after it has been put into use is known as
a) Software modification
b) Software change
c) Software reengineering
d) Software maintenance

55. If a system uses memory-mapped I/O, how much addressable memory is available if the address bus is 16 bits and 256 I/O addresses are reserved?
a) 65,536 bytes b) 65,280 bytes
c) 32,768 bytes d) 32,512 bytes

56. Calculate the effective address for the instruction 'LOAD R1, 100(R2)' if the content of R2 is 200
a) 100 b) 200 c) 300 d) 400

57. Which of the following is an example of an I/O control method where the CPU polls each device?
a) Interrupt-driven I/O
b) Direct Memory Access (DMA)
c) Programmed I/O
d) Memory-mapped I/O

58. Suppose you have coins of denominations 1, 3 and 4. You use a greedy algorithm, in which you choose the largest denomination coin which is not greater than the remaining sum. For which of the following sums, will the algorithm NOT produce an optimal answer?
a) 20 b) 6 c) 12 d) 5

59. The worst case time complexity of quick sort is ______
a) O(n) b) $O(n^2)$
c) O(n log n) d) O(log n)

60. Which of the following algorithm is not in-place?
a) Selection sort b) Quick sort
c) Merge Sort d) Bubble Sort

61. Predict the output of following program

```
#include <stdio.h>
int fun(int n)
{
   if (n == 4)
      return n;
   else return 2*fun(n+1);
}
int main()
{
  printf("%d", fun(2));
  return 0;
}
```

a) 4 b) 8 c) 16 d) Runtime Error

62. A Binary search tree is generated by inserting the following elements in order,
50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24
The number of nodes in the left sub tree and right sub tree of the root is ________
a) (4, 7) b) (7, 4) c) (8, 3) d) (3, 8)

63. A variant of the linked list in which none of the node contains NULL pointer is_______
a) Singly linked list b) Doubly linked list
c) Circular linked list d) None

64. If p and q are independent propositions, which of these is always false?
a) $p \wedge \neg q$ b) $\neg p \wedge \neg q$ c) $p \rightarrow q$ d) $p \leftrightarrow \neg q$

65. How many equivalence relations can be defined on a set of size 3?
a) 5 b) 10 c) 15 d) 6

66. In a 7-node directed cyclic graph, the number of Hamiltonian cycle is to be
a) 180 b) 720 c) 360 d) 540

67. What is the purpose of a Use Case Diagram in UML?
a) To model the flow of data within the system
b) To show the system's interactions with external entities
c) To define the class structure of the system
d) To represent the system's behavior over time

68. In the Rational Unified Process (RUP), which phase focuses on creating the architectural foundation of the system?
a) Inception
b) Elaboration
c) Construction
d) Transition

69. Which type of testing ensures that previously developed and tested software still works after changes?
a) Regression Testing
b) Acceptance Testing
c) Integration Testing
d) System Testing

70. Which of the following acts as a temporary storage location to hold an intermediate results for mathematical and logical calculations
a) RAM
b) Accumulator
c) Program counter
d) Memory Address Register

71. What is the main disadvantage of the **SSTF (Shortest Seek Time First)** disk scheduling algorithm?
a) It increases seek time compared to FCFS
b) It causes starvation for distant requests
c) It requires additional hardware support
d) It is less efficient than SCAN

72. Which of the following replacement policies suffers from Belady's anomaly?
a) LRU
b) FIFO
c) Optimal
d) NRU

73. Thrashing occurs when :
a) The CPU is overloaded
b) Processes spend more time paging than executing
c) All processes are in a ready state
d) There is insufficient CPU scheduling

74. Which protocol type does Wi-Fi use for media access control?
a) FDMA
b) CDMA
c) CSMA/CA
d) TDMA

75. In Ethernet addressing, if all the bits are 1s, the address is ____________
a) Unicast
b) Multicast
c) Broadcast
d) None of these

76. Standard Ethernet uses_________ encoding
a) AMI
b) Manchester
c) Differential Manchester
d) NRZ

77. Which phase of the Unified Process involves deploying the system to its users?
a) Inception
b) Elaboration
c) Construction
d) Transition

78. In a weather monitoring system the **Observer** pattern is implemented. The system has the following components:

A **WeatherStation** object (subject) that notifies its observers whenever the temperature changes.

There are 200 **Display** objects (observers) that are subscribed to the **WeatherStation** to receive updates.

If each observer object stores the temperature (which is an integer) and has additional overhead of 4 bytes per observer for other attributes, calculate the total memory required to store all observer objects

a) 800 bytes b) 1200 bytes c) 1600 bytes d) 2000 bytes

79. ________________ Design pattern converts the interface of a class into another interface clients expect

a) Bridge method
b) Adapter method
c) Proxy method
d) Decorator method

80. Which of the following is a valid SQL aggregate function?
a) COUNT b) AVG c) MAX d) All of these.

81. Which layer in the OSI model is responsible for data presentation?
a) Network layer
b) Session layer
c) Presentation layer
d) Transport layer

82. The purpose of object-oriented analysis is to_______
a) Identify the objects and their attributes
b) Define the system requirements
c) Model the system using object-oriented concepts
d) All of these

83. Use case diagram is used to __________
a) Model the interactions between users and the system
b) Model the classes and their relationships in the system
c) Model the flow of activities in the system
d) None of these

84. The purpose of the factory method design pattern is to _______
a) To define an interface for creating objects without specifying their concrete classes
b) To encapsulate object creation
c) To manage object destruction
d) To provide a way for multiple inheritance

85. Which of the following is TRUE?
a) Every relation in 3NF is also in BCNF
b) A relation R is in 3NF if every non-prime attribute of R is fully functionally dependent on every key of R
c) Every relation in BCNF is also in 3NF
d) No relation can be in both BCNF and 3NF

86. The five items: A, B, C, D, and E are pushed in a stack, one after other starting from A. The stack is popped four times and each element is inserted in a queue. The two elements are deleted from the queue and pushed back on the stack. Now one item is popped from the stack. The popped item is__________
a) A b) B c) C d) D

87. If an array is declared as int arr[5][5], how many elements can it store?
a) 5 b) 25 c) 10 d) 0

88. Which of the binary tree traversal requires a Queue?
a) Level-order b) pre-order c) post-order d) in-order

89. If R is a relation on set A= {1, 2, 3, 4} defined by R= {(x,y) | x+y=5}, then which pairs belong to R?
a) {(1,4),(2,3),(3,2),(4,1)}
b) {(1, 4), (2, 3), (4, 1)}
c) {(1, 4), (2, 3), (3, 2)}
d) {(1, 3), (2, 3), (4, 1)}

90. What is the Cartesian product of set A and set B, if the set A = {1, 2} and set B = {a, b}?
a) { (1, a), (1, b), (2, a), (b, b) }
b) { (1, 1), (2, 2), (a, a), (b, b) }
c) { (1, a), (2, a), (1, b), (2, b) }
d) { (1, 1), (a, a), (2, a), (1, b) }

91. The unit which acts as an intermediate agent between memory and backing store to reduce process time is ___________
a) Translation Lookaside Buffer's
b) Registers
c) Page Tables
d) Cache

92. The bit used to signify that the cache location is updated is ________
a) Flag bit b) Reference bit
c) Update bit d) Dirty bit

93. Which of the following algorithms is used to find the articulation points in a graph?
a) Bellman-Ford algorithm
b) Floyd-Warshall algorithm
c) Depth First Search
d) Kruskal's algorithm

94. In the N-Queens problem, what does backtracking do when a placed queen results in no valid positions for the next queen?
a) Stops the program
b) Recalculates all possible placements for the queens
c) Removes the last placed queen and tries the next possibility
d) Ignores the invalid position and moves forward

95. Which data structures are used to formulate the Breadth-first Search (BFS)?
a) Stack
b) Queue
c) Heap
d) Linked List

96. Which strategy is used to handle deadlocks by ensuring that at least one of the necessary conditions for deadlock does not hold?
a) Deadlock Avoidance
b) Deadlock Detection and Recovery
c) Deadlock Prevention
d) Process Termination

97. Which disk scheduling algorithm minimizes the total seek time?
a) First-Come, First-Served (FCFS)
b) Shortest Seek Time First (SSTF)
c) Elevator (SCAN)
d) Circular SCAN (C-SCAN)

98. Access matrix model for user authentication contains ___________
a) A list of objects
b) A list of domains
c) A function which returns an object's type
d) All of these

99. A system digitizes an analog signal with a maximum frequency of 4 kHz. According to the Nyquist theorem, what is the minimum sampling rate required to avoid aliasing?
a) 4 kHz
b) 8 kHz
c) 16 kHz
d) 32 kHz

100. Using Go-Back-N ARQ with a window size of 4, how many frames can be sent before an acknowledgment is required?
a) 1 frame
b) 4 frames
c) 5 frames
d) 8 frames

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